

High-Speed Signal Integrity Test Board

Measurement Report — Channel Loss, Ground Bounce & Crosstalk

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1. Project Overview

This report documents empirical signal integrity measurements taken from a custom high-speed digital test board, designed and routed on standard FR4 substrate to characterize signal degradation at 20 Mbps. SMA connectors and buffer/inverter logic gates were used to isolate and independently evaluate three transmission-line effects: channel (insertion) loss, ground bounce, and trace-to-trace crosstalk. All measurements were captured on Keysight InfiniiVision oscilloscopes (DSOX3052G, 500 MHz / 5 GSa/s, unless otherwise noted).

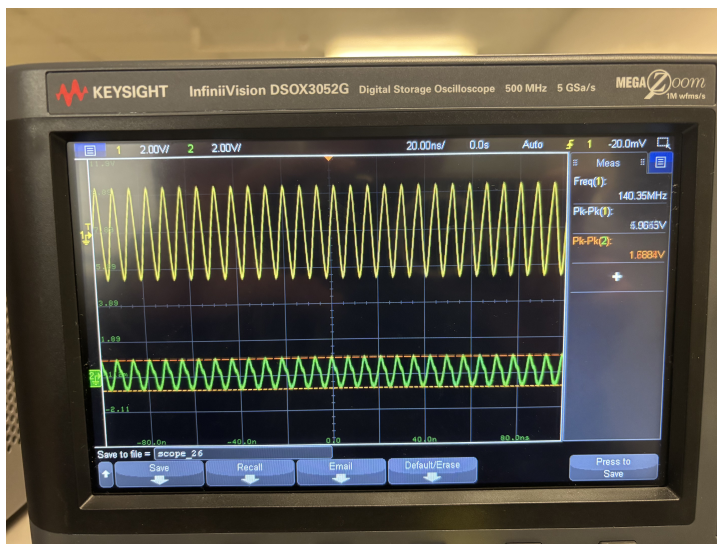
Test methodology, per the project team's presentation:

- Channel loss: trace routing was designed with wire spacing at least $3\times$ the trace width to avoid coupling contamination of the loss measurement.
- Crosstalk: the aggressor trace was always driven from its near-end SMA, with its far-end terminated in $50\ \Omega$. Near-end crosstalk (NEXT) was measured at the victim's near-end SMA (far end terminated $50\ \Omega$); far-end crosstalk (FEXT) was measured at the victim's far-end SMA (near end terminated $50\ \Omega$).
- Ground bounce: measurements were taken both with and without a decoupling capacitor ($4.7\ \mu\text{F}$) at the IC's ground pin, to evaluate its effect on suppressing the bounce.

2. Channel Loss

Insertion loss was measured by comparing the peak-to-peak input signal (CH1) against the output after the channel (CH2), computed as $20\cdot\log_{10}(V_{\text{out}} / V_{\text{in}})$. With the mislabeled crosstalk points removed, the remaining data shows a smooth, physically sensible increase in loss with frequency.

Sample	Freq	Vin (Pk-Pk)	Vout (Pk-Pk)	Insertion Loss
IMG_8273	not captured	261.3 mV	253.8 mV	-0.25 dB
IMG_8275	not captured	3.41 V	1.76 V	-5.74 dB
IMG_8279	127.11 MHz	3.54 V	1.93 V	-5.26 dB
IMG_8297	158.63 MHz	1.39 V	480 mV	-9.26 dB
IMG_8298	307.71 MHz	2.58 V	344 mV	-17.5 dB



CH1 input (yellow) vs. CH2 output (green) — channel loss at 140.35 MHz.

3. Ground Bounce

Ground bounce was measured as noise coupled onto the ground reference during signal switching. Two configurations were tested at 1 MHz and 10 MHz: with and without a 4.7 μ F decoupling capacitor at the IC's ground pin. A few earlier single-edge captures (different scope, configuration not specified in the presentation) are included for reference.

Sample	Freq / Config	Signal Pk-Pk	Bounce Pk-Pk	Bounce % of Signal
IMG_8316	1 MHz — with capacitor	3.9 V	76 mV	1.95%
IMG_8315	10 MHz — with capacitor	3.38 V	80.40 mV	2.38%
IMG_8317	1 MHz — without capacitor	3.8 V	111 mV	2.92%
IMG_8314	10 MHz — without capacitor	3.30 V	110.55 mV	3.35%
IMG_8306	EDUX1052G, single edge	9.6 V	1.0 V	~10.4%
IMG_8307	DSOX3052G, ringdown	— (detail view)	79.7 mV	—



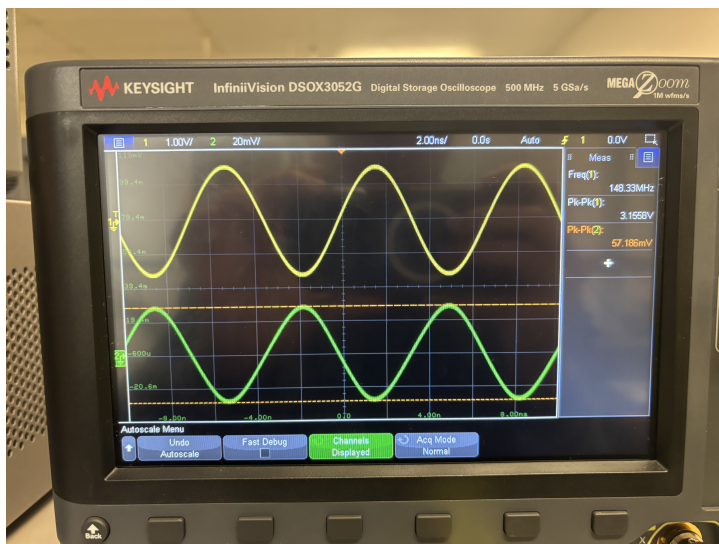
Ground bounce at 10 MHz, without decoupling capacitor: CH1 switching signal (yellow) vs. CH2 bounce (green).

4. Crosstalk

Crosstalk was measured with the aggressor trace driven from its near-end SMA (far end terminated 50 Ω). Near-end crosstalk (NEXT) was read at the victim's near-end SMA; far-end crosstalk (FEXT) was read at the victim's far-end SMA. Coupling % = $V_{\text{victim}} / V_{\text{aggressor}}$.

Near-End Crosstalk (NEXT)

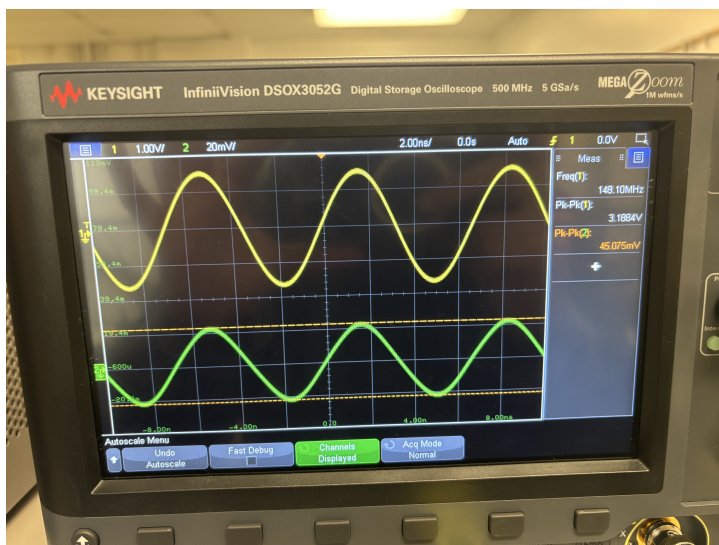
Sample	Freq	Aggressor Pk-Pk	Victim Pk-Pk	Coupling %
IMG_8310	999.6 kHz	8.0 V	68 mV	0.85%
IMG_8311	10.04 MHz	8.2 V	68 mV	0.83%
IMG_8301	148.33 MHz	3.16 V	57.2 mV	1.81%
IMG_8299	150 MHz (repeat)	3.30 V	58 mV	1.76%
IMG_8300	348.18 MHz	2.27 V	68.0 mV	3.00%



NEXT at 148 MHz: CH1 aggressor (yellow) vs. CH2 near-end victim coupling (green).

Far-End Crosstalk (FEXT)

Sample	Freq	Aggressor Pk-Pk	Victim Pk-Pk	Coupling %
IMG_8312	1.0000 MHz	8.2 V	51 mV	0.62%
IMG_8313	10.00 MHz	8.4 V	52 mV	0.62%
IMG_8302	148.10 MHz	3.19 V	45.1 mV	1.41%
IMG_8303	348.96 MHz	2.31 V	54.9 mV	2.38%



FEXT at 148 MHz: CH1 aggressor (yellow) vs. CH2 far-end victim coupling (green).

6. Conclusions

[Space reserved for your analysis — e.g., how measured channel loss compares to theoretical transmission-line predictions at 20 Mbps, why NEXT consistently exceeds FEXT, and how effective the decoupling capacitor was at suppressing ground bounce.]